

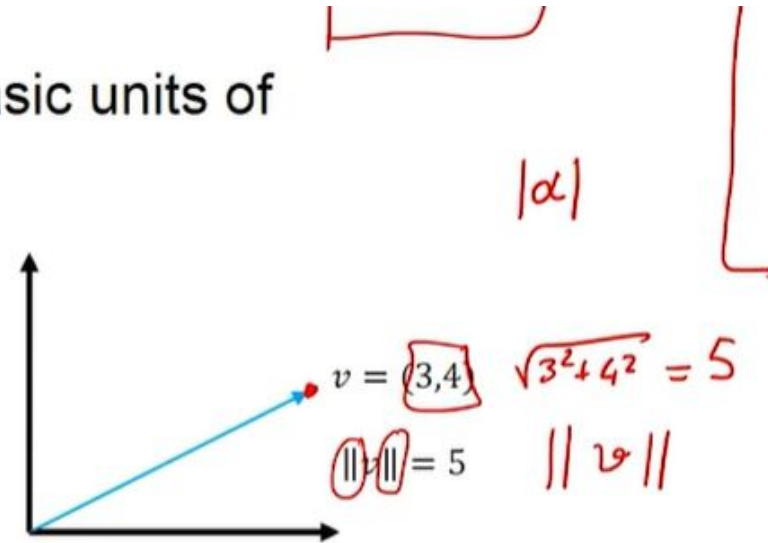
Norms

- Machine Learning uses tensors as the basic units of representation

- Vectors, Matrices, etc..

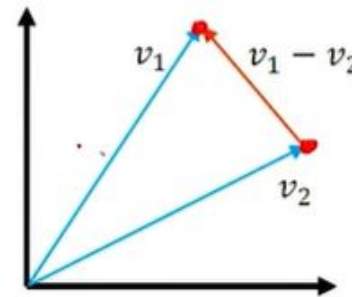
- Two reasons to use norms

1. To estimate how “big” a vector/tensor is

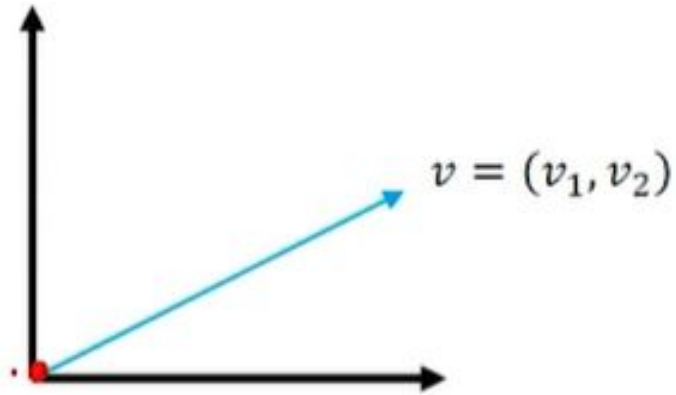


2. To estimate “how close” one tensor is to another

- That is how “big” the *difference between two tensors* is



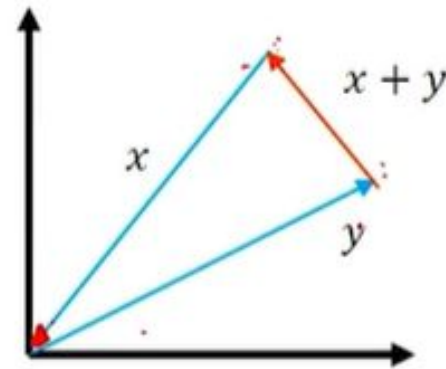
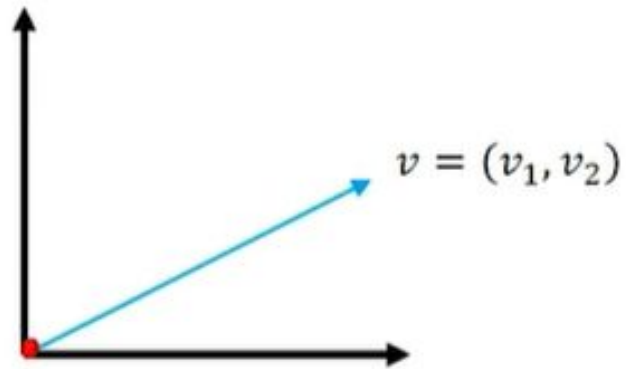
Norms are a way of measuring the “length” of vectors, matrices, etc



Mathematically, a norm is any function $\mathbf{f}$ that satisfies

- $f(\mathbf{x}) = 0 \Rightarrow \mathbf{x} = \mathbf{0}$

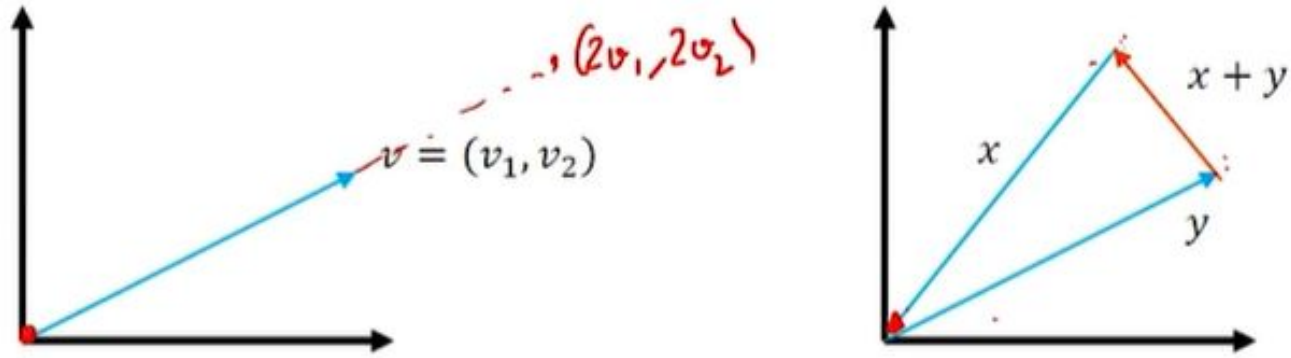
Norms are a way of measuring the “length” of vectors, matrices, etc



Mathematically, a norm is any function f that satisfies

- $f(x) = 0 \Rightarrow x = 0$
- $f(x + y) \leq f(x) + f(y)$ (Triangle Inequality)

Norms are a way of measuring the “length” of vectors, matrices, etc



Norm

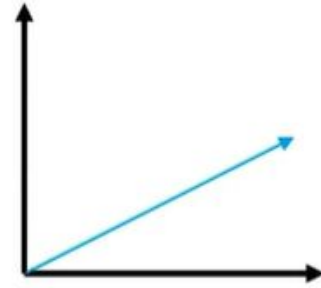
Mathematically, a norm is any function f that satisfies

- $f(x) = 0 \Rightarrow x = 0$
- $f(x + y) \leq f(x) + f(y)$ (Triangle Inequality)
- $\forall \alpha \in \mathbb{R}, f(\alpha x) = |\alpha|f(x)$ (Linearity)

Some standard norms

$$f(\vec{v}) \rightarrow \text{Scalar (tra)} \quad \vec{v} = \begin{bmatrix} -5 \\ 3 \\ 2 \end{bmatrix}$$

- $f(\mathbf{x}) = 0 \Rightarrow \mathbf{x} = \mathbf{0}$ ✓
- $f(\mathbf{x} + \mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$ (Triangle Inequality) ✓
- $\forall \alpha \in \mathbb{R}, f(\alpha \mathbf{x}) = |\alpha|f(\mathbf{x})$ (Linearity) ✓



```
>> v = [-5, 3, 2]'  
  
v =  
  
-5  
 3  
 2
```

Vector Norms

1. **Euclidean Norm** : $\|v\|_2 = (v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2)^{\frac{1}{2}}$

- Also called the 2-norm or the L^2 norm
- Corresponds to our usual notion of distance

2. **1-norm** : $\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$

3. **p-Norm** : $\|v\|_p = (|v_1|^p + |v_2|^p + \dots + |v_n|^p)^{\frac{1}{p}}$

4. **∞ -Norm** : $\|v\|_\infty = \max(|v_1|, |v_2|, \dots, |v_n|)$
max-Norm

■ **Matrices : Frobenius norm** $A_F = (\sum_{i,j} A_{ij}^2)^{\frac{1}{2}}$

```
>> A = [1 2; 2 0]  
  
A =  
  
 1  2  
 2  0
```

```
>> norm(A, 'fro')
```

ans =

3